Lesson 14 Teacher's Key: Raising the Stakes

Godot Engine - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

Same idea as every engine this week - replace hardcoded literals with wave-dependent formulas - expressed as GDScript expressions inside the spawn_wave() function this course has been building since Lesson 10.

Level 1 - Full Worked Solution

- range(4 + wave) and enemies_remaining = 4 + wave must use the identical expression - any mismatch is exactly the kind of bug this Key's troubleshooting section calls out.
- No new variables are needed for Level 1 - this is purely a formula change inside a function that already exists.
- Wave 1 still spawns exactly 5 (4 + 1 = 5), so nothing about the game's opening feel changes - only later waves escalate.

Why This Structure

- Keeping both the loop bound and enemies_remaining's value inside the SAME function (spawn_wave()) rather than split across files makes the 'one source of truth' discipline easy to see and easy to verify at a glance.
- For Level 2, setting enemy.speed on each freshly-instantiated enemy (rather than changing enemy.tscn's own default) keeps the scaling logic entirely inside spawn_wave() - the enemy scene itself stays exactly as simple as Lesson 7 left it.

Level 2 / Level 3 Talking Points

- Ask students to trace through spawn_wave() for wave 3 by hand: range(4+3)=range(7), enemies_remaining=7 - a good comprehension check for reading a formula inside real code.
- For students comparing engines: point out that Unreal's identical (4 + Wave) formula, expressed as Blueprint math nodes instead of GDScript, is conceptually the exact same idea - a good cross-engine parallel.

Common Student Mistakes

- Updating the loop's range() but forgetting to update enemies_remaining (or vice versa).

- Setting enemy.speed BEFORE add_child() instead of after - harmless in Godot 4 either way, but worth confirming the assignment target is actually the enemy just created, not a stale reference.
- Leaving enemy.tscn's default speed := 60.0 as the ONLY place speed is set, forgetting the new per-instance override in spawn_wave() entirely.